

HOMESERVANT

Backend Developer Guide

How the HomeServant API is built

Version of 27 September 2026. Written from the code in the HomeServant-MVP repository.

Source: `docs/guides/backend_guide.html` — rebuild with `python3 docs/guides/build.py`.

Contents

1. What the backend is and what it runs on
2. Folder and module layout
3. How a request is handled
4. Authentication and sessions
5. Roles, admin levels and permissions
6. The data model
7. Business flows (bookings, payments, marketplace, chat)
8. Real-time: the WebSocket gateway
9. Background jobs
0. Files, email and other outside services
1. Configuration (environment variables)
2. Running, migrating and deploying
3. API reference
4. Conventions for new code

1. What the backend is and what it runs on

The HomeServant API is the single source of truth for users, listings, bookings, payments, the marketplace, chat, notifications and the admin console. The Flutter app (web and mobile) is only a client of it.

Piece	Choice and why
Framework	NestJS 10 (TypeScript). Its modules, guards and dependency injection keep auth and permission rules in one place.
Database	PostgreSQL on Supabase , accessed through Prisma 5 . Prisma gives a typed schema, generated queries and versioned migrations.
Real-time	Socket.IO , through Nest's WebSocket gateway, for chat, notifications and live admin updates.
File storage	Supabase Storage (public bucket). The client uploads directly using one-time signed links.
Payments	Paystack : charges, webhooks, transfers and refunds, in an escrow model.
Email	Resend in production; the console in development.
Hosting	Render (Docker web service, <code>render.yaml</code>). Migrations run automatically at start-up.

All routes are served under the `/api` prefix, for example `https://homeservant-api-c8w7.onrender.com/api/health`.

2. Folder and module layout

Everything is in `backend/`. Each feature is a Nest *module* in `src/<name>/`, usually containing a `*.controller.ts` (HTTP routes), a `*.service.ts` (business logic), a `dto/` folder (validated request bodies) and a `*.module.ts`.

Module	Responsibility
auth	Signup and verification, login, Google, 2FA, JWT issuing and refresh, password reset and change, deactivate and delete, cleanup of deactivated accounts.
users	Own profile (<code>/users/me</code>), bank details, the <code>profileCompletedAt</code> flag.
properties	Listings: browse, create, update, delete.
bookings	Rentals and shortlets: requests, payment, inspection, move-in, refund, reject, renew, tenancy agreement; the lease life-cycle job.
payments	The single home for escrow money movement (charge, hold, release, refund).
paystack	Paystack HTTP client, bank list and account-name lookup, webhook route.
favorites , reviews , reports	Wishlist, reviews (only from real renters), user reports worked by admins.
chat	Threads and messages, support chats, claim, transfer, reassign, access rules, the WebSocket gateway, online presence, support alerts and cleanup jobs.
notifications	Notification rows plus live pushes (the only place notifications are created).
vendors , marketplace-products , marketplace-orders	Shops (applications and approval), products, orders and items, shipping and tracking, auto-release job.
delivery	Courier integration behind an interface: console (default) or GIG Logistics (scaffolding only).
storage	Signed upload links; checks that a URL points to one of our own images.
otp , mail	One-time codes, and email through a pluggable provider (console or Resend).
admin	Everything behind the admin console, plus the one-time bootstrap route.
activity-log	Audit trail of sensitive actions.
common	Guards, decorators and the error filter shared by all modules.
prisma , health	Database client; <code>GET /api/health</code> for Render's health check.

3. How a request is handled

Set up in `src/main.ts` and `src/app.module.ts` :

1. **Helmet** (security headers) and **compression**. `trust proxy` is on, so client IPs are correct behind Render's proxy.
2. **CORS**: only the origins listed in `CORS_ORIGINS` can call the API from a browser.
3. **Rate limiting** (`ThrottlerGuard` , global): 100 requests per minute per IP by default, and stricter on sensitive routes (for example the bootstrap route allows only 5 per minute).
4. **Guards** on each route decide who may call it (section 5).
5. **ValidationPipe**: request bodies are checked against their DTO classes. Unknown fields are rejected (`whitelist` and `forbidNonWhitelisted`).
6. The controller calls a service, which reads and writes the database through Prisma.
7. **HttpExceptionHandler** turns every error into `{ statusCode, message, error }`. The app shows `message` directly, so write messages for end users.

4. Authentication and sessions

Tokens

- **Access token:** a JWT signed with `JWT_ACCESS_SECRET`, valid for 15 minutes (`JWT_ACCESS_TTL`). It's sent as `Authorization: Bearer ...`.
- **Refresh token:** a JWT signed with `JWT_REFRESH_SECRET`, valid for 30 days (`JWT_REFRESH_TTL`). Only a hash of it is stored in `RefreshToken`. **Each one can be used once:** `POST /auth/refresh` revokes it and issues a new pair.
- Logging out, resetting or changing a password, deactivating or deleting all revoke refresh tokens.
- Changing either secret signs everyone out immediately. The Admin Guide's emergency procedure relies on this.

Sign-up and login flows

- **Signup:** `POST /auth/signup` creates an unverified user and emails a code; `POST /auth/verify-signup` verifies it and returns tokens. Signing up again with an email that was never verified takes over the unverified record and sends a fresh code.
- **Login:** `POST /auth/login` can return:
 - tokens;
 - `requiresTwoFactor` (a code was emailed; finish with `/auth/verify-2fa`);
 - `requiresReactivation` (the account is deactivated; send the same request again with `reactivate: true`);
 - `requiresEmailVerification` (the email was never verified; a fresh signup code was sent).
- **Google:** `POST /auth/google` verifies Google's ID token against `GOOGLE_CLIENT_ID`, then finds or creates the user. An existing email is linked to the Google account.
- **Profile completion:** `User.profileCompletedAt` is set the first time a profile has every field signup requires, and is never cleared. The app's "finish your profile" redirect relies only on this flag, which is included in every auth response.
- **Password reset** is by emailed code (`/auth/forgot-password` then `/auth/reset-password`). It's refused for admin accounts: another admin must reset those.

5. Roles, admin levels and permissions

`User.role` is one of `TENANT`, `LANDLORD`, `VENDOR` or `ADMIN`. Admins also have `adminLevel`: `SUPPORT` < `MODERATOR` < `SUPER_ADMIN`.

Guard / decorator	What it does
<code>JwtAuthGuard</code>	Requires a valid access token and sets <code>request.user</code> .
<code>RolesGuard</code> + <code>@Roles(...)</code>	Requires one of the listed roles. For admin routes it re-reads role and level from the database on every request , so demoting or removing an admin takes effect on their very next request.
<code>AdminLevelGuard</code> + <code>@MinAdminLevel(...)</code>	Requires at least that admin level.
<code>MustChangePasswordGuard</code>	Blocks an admin still on a temporary password (except routes marked <code>@AllowMustChangePassword</code>).

Some rules are more specific than a minimum level, so services check them themselves:

- **Admin password resets:** you can't reset the password of an admin ranked above you.
- **Last super admin:** the last super admin can't be demoted or removed.
- **Support chat access** (`chat/support-access.ts` and `ChatService.canWrite / assertCanRead`):
 - non-participant admins can reach a support thread only while it's unclaimed;
 - a `SUPER_ADMIN` can read, but not write to, threads another admin is handling;
 - only a super admin can reassign.

The full admin permission table is in the Admin Guide.

6. The data model

Defined in `prisma/schema.prisma` (read its comments: most fields explain why they exist). Main tables:

Model	What it stores
User	Every account: role, admin level, profile fields, bank details, 2FA, verification, deactivation, <code>profileCompletedAt</code> , <code>adminOnDuty</code> , referral code.
<code>RefreshToken</code> , <code>OtpCode</code> , <code>PendingAdmin</code>	Sessions; one-time codes (10-minute lifetime); admin invites not yet confirmed.
<code>Property</code> , <code>Favorite</code> , <code>Review</code>	Listings (category, price unit, photos, video, occupancy, messaging toggle), wishlists, reviews.
<code>Booking</code> , <code>TenancyAgreement</code>	Rentals and shortlets with their status and lease dates; the agreement record.
Payment	The escrow ledger: one row per charge, for either a booking or an order item.
<code>VendorProfile</code> , <code>Product</code> , <code>MarketplaceOrder</code> , <code>MarketplaceOrderItem</code>	Shops, products, orders (one payment per item).
<code>Thread</code> , <code>ThreadParticipant</code> , <code>Message</code> , <code>ThreadTransferLog</code>	Chats, including support threads (<code>isSupport</code> , <code>status</code> , <code>assignedAdminId</code> , reminder timestamps) and their transfer history.
<code>Notification</code>	In-app notifications, with an optional <code>threadId</code> link.
<code>Report</code> , <code>ActivityLog</code>	User reports; the admin audit trail.

When an account is deleted, a sender's messages are kept but lose their sender (`onDelete: SetNull`); the same happens to activity-log actors and transfer records. Code that reads these must allow for a missing person.

7. Business flows

Bookings

- **Shortlet:** the tenant requests dates (`PENDING`). The landlord accepts or declines (`ACCEPTED` / `DECLINED`). The tenant pays (`PAID`), and payment is released to the landlord.
- **Rental (house, apartment, self-con):** creating the booking charges straight away (`PENDING` , then `PAID_AWAITING_INSPECTION` once paid). An inspection date is proposed and confirmed (`INSPECTION_PROPOSED` , `INSPECTION_CONFIRMED`). Moving in (`MOVED_IN`) releases the payment. Before move-in, the tenant can get a refund (`REFUNDED`) or the landlord can reject. A renewal charges again, near the end of the lease.

Payments (escrow)

All in `PaymentsService` . HomeServant charges the full amount through Paystack and holds it in its own balance (`PAID_HELD`). Later it transfers the recipient's share to their verified bank account (`RELEASED`) or refunds the payer (`REFUNDED`).

- The platform keeps **5%** on a release, or **0.2%** on a rental refunded before move-in. Fees are in basis points, and amounts are converted to kobo only at the Paystack boundary.
- Paystack confirms payments at `POST /api/paystack/webhook` . The signature (`x-paystack-signature`) is checked against the raw request body before anything is trusted.

Marketplace

- An order is split into items, one payment each. The vendor ships (a delivery item gets a tracking record), the buyer confirms receipt, and the vendor's share is released.
- If the buyer never confirms, a daily job releases the payment 7 days after shipping (or 7 days after payment for pickup items), unless there's an open report about that product.
- Shops are opened by tenant accounts, must be approved by an admin, and a suspended shop's products are hidden.

Chat and support

- Normal threads link two users, optionally about a property or an order, and are reused rather than duplicated.
- Tenant/landlord messaging is tied to payment (`ChatService.landlordTenantBlockReason`), so both sides do business on the platform. A paid booking (`PAID` through `MOVED_IN`) opens messaging both ways. If there is no paid booking and a booking was refunded (the tenant's own refund, or the landlord rejecting a paid booking), messaging is closed *both ways* until the tenant pays again. With nothing paid yet, a landlord may reach out and the tenant may only answer a thread the landlord has already written in. Bookings count if they are on the thread's property, or on any of that landlord's properties for a thread not tied to a listing. The rule is checked when a thread is started and on every message; an existing thread still opens so its history stays readable, and `GET /threads/:id/summary` returns `lockedReason` , which the app shows in place of the composer. Support and marketplace order threads are not affected.
- When messaging closes on a refund, `PaymentsService` posts a `SYSTEM` message (no sender) through `ChatService.postBookingSystemMessage` so both sides see why in the chat. A tenant's

own refund adds it only to an existing conversation; a landlord's rejection starts one if needed, and the tenant's "Booking rejected" notification links to that chat.

- A support thread starts with just the user. It sits in the shared queue until an admin claims it, either by opening it (`PATCH /threads/:id/claim`) or by replying, which is race-safe. It can then be transferred, reassigned by a super admin, or resolved.
- Notifications for unclaimed threads are collapsed into one per admin (`NotificationsService.upsertThreadAlert`) and cleared for everyone else on claim.
- Admins marked Away get them silently (`User.adminOnDuty`).
- Resolved support threads are removed after 30 days.

Notifications

`NotificationsService.create` is the only place notifications are created. It stores the row and pushes `notification:new` to the user's live connection, with a transient `silent` flag when the app shouldn't play a sound or show a banner.

8. Real-time: the WebSocket gateway

`chat/chat.gateway.ts` . Clients connect with `auth: { token: <access token> }` . Each socket joins:

- `user:<id>` (its own room);
- `admins` , if it's an admin;
- `thread:<id>` , if it asks and is allowed.

Event (server to client)	When
<code>message:new</code>	A message is sent. It goes to the other participants, and to all admins while a support thread is unclaimed.
<code>message:read</code>	The other side read the thread.
<code>notification:new</code>	A notification was created or updated (may be <code>silent</code>).
<code>thread:claimed</code>	A support thread changed hands: claimed, transferred or reassigned.
<code>thread:access-revoked</code>	This socket was removed from a thread room it's no longer allowed in.
<code>admin:badges-changed</code>	An admin badge count changed.

Online presence (`PresenceService`) tracks connected sockets per user and records `lastActiveAt` .

9. Background jobs

Scheduled with `@nestjs/schedule`. If Render's free plan has the server asleep, a job runs on the first tick after it wakes. Every job is written to catch up safely.

Job	When	What
<code>SupportAlertsService</code>	Every minute	Unclaimed support chat older than 5 minutes: one reminder to all admins. Customer unanswered for 10 minutes: one alert to super admins.
<code>AccountCleanupService</code>	Daily, 03:00	Permanently deletes accounts deactivated more than 30 days ago (skipping ones with active leases or orders).
<code>SupportChatCleanupService</code>	Daily, 03:00	Deletes support messages older than 30 days, and emptied threads.
<code>LeaseLifecycleService</code>	Daily, 04:00	Re-lists properties whose lease ended without renewal; sends rent-expiry reminders 30, 15 and 0 days before.
<code>MarketplaceAutoReleaseService</code>	Daily, 05:00	Releases held marketplace payments after 7 days, unless disputed.

10. Files, email and other outside services

- **Uploads:** `POST /uploads/sign` takes a file name and folder. Only image extensions and known folders are accepted. It returns a one-time Supabase upload link and the file's public URL, stored under `<folder>/<userId>/<random>`. Any endpoint that saves an image URL first calls `StorageService.assertIsOwnImage`, which checks it's in our bucket and really an image.
- **Email and codes:** `OTP_PROVIDER` selects `console` (codes printed to the server log, for development) or `resend` (real email). `MailService` uses the same setting.
- **Paystack:** bank list, account-name lookup, charges, transfer recipients, transfers, refunds, webhook.
- **Google:** ID tokens are verified with Google's library against `GOOGLE_CLIENT_ID`.
- **Delivery:** `DELIVERY_PROVIDER` selects `console` (default) or `gig`. The GIG integration is unverified scaffolding.

11. Configuration (environment variables)

Set in Render (see `render.yaml`) and locally in `backend/.env` (copy `.env.example`).

Variable	Purpose
<code>DATABASE_URL</code> , <code>DIRECT_URL</code>	Supabase Postgres: the pooled connection for the app, the direct one for migrations.
<code>SUPABASE_URL</code> , <code>SUPABASE_SERVICE_ROLE_KEY</code> , <code>SUPABASE_STORAGE_BUCKET</code>	Storage access. The service role key must never reach the client.
<code>JWT_ACCESS_SECRET</code> , <code>JWT_REFRESH_SECRET</code>	Token signing secrets. Changing them signs everyone out.
<code>JWT_ACCESS_TTL</code> , <code>JWT_REFRESH_TTL</code>	Token lifetimes (default <code>15m</code> and <code>30d</code>).
<code>OTP_PROVIDER</code> , <code>RESEND_API_KEY</code> , <code>RESEND_FROM_EMAIL</code>	Email and code delivery.
<code>GOOGLE_CLIENT_ID</code>	Google web OAuth client; must match the app's.
<code>PAYSTACK_SECRET_KEY</code>	Paystack API and webhook signature checks.
<code>ADMIN_BOOTSTRAP_SECRET</code>	Unlocks the one-time "create first super admin" route. It refuses once any admin exists.
<code>CORS_ORIGINS</code>	Comma-separated web origins allowed to call the API.
<code>SUPPORT_UNCLAIMED_REMINDER_MINUTES</code> , <code>SUPPORT_REPLY_ESCALATION_MINUTES</code>	Support reminder delays (defaults 5 and 10).
<code>DELIVERY_PROVIDER</code> , <code>GIG_API_KEY</code> , <code>GIG_API_BASE_URL</code>	Courier integration.
<code>PORT</code>	HTTP port (Render uses 3000).

12. Running, migrating and deploying

Local

```
cd backend
cp .env.example .env          # fill in Supabase + two random JWT secrets
npm install
npm run prisma:migrate        # create/upgrade the local schema
npm run start:dev             # http://localhost:3000/api, reloads on change
npx tsc --noEmit -p tsconfig.json # type-check
```

With `OTP_PROVIDER=console`, verification codes are printed in this terminal.

Changing the database

1. Edit `prisma/schema.prisma`, with a comment explaining any new field.
2. Run `npx prisma migrate dev --name what_changed`, or hand-write `prisma/migrations/<timestamp>_<name>/migration.sql`. Prefer additive changes (new nullable columns, or columns with defaults) so the live app keeps working during deploys.
3. Commit the migration folder with the code.

Deploying

Render builds the Docker image. On start the container runs `npx prisma migrate deploy` (applying any new migrations) and then `node dist/main.js`. Deploy the backend **before or together with** any web release that depends on new fields.

13. API reference

All paths are under `/api`. "Auth" means a valid access token; "Admin (level)" means role ADMIN with at least that level.

Route	Access
GET <code>/health</code>	Public
POST <code>/auth/signup</code> <code>verify-signup</code> <code>login</code> <code>google</code> <code>verify-2fa</code> <code>resend-otp</code> <code>refresh</code> <code>forgot-password</code> <code>reset-password</code>	Public (rate-limited)
POST <code>/auth/logout</code> , PATCH <code>/auth/password</code> , PATCH <code>/auth/deactivate</code> , DELETE <code>/auth/me</code>	Auth
GET PATCH <code>/users/me</code> , PATCH <code>/users/me/bank-details</code>	Auth
GET <code>/properties</code> , GET <code>/properties/:id</code>	Public
POST <code>/properties</code> , PATCH DELETE <code>/properties/:id</code>	Landlord (owner of the listing)
POST <code>/bookings</code> , GET <code>/bookings/mine landlord</code> , PATCH <code>/bookings/:id/respond</code> , POST <code>/bookings/:id/pay inspection reject moved-in refund renew</code> , PATCH <code>/bookings/:id/inspection/respond</code>	Auth (tenant or landlord, checked per booking)
GET <code>/favorites</code> , POST <code>/favorites/:propertyId/toggle</code>	Auth
GET <code>/reviews</code> , GET <code>/reviews/mine</code> , POST <code>/reviews</code> , DELETE <code>/reviews/:id</code>	Public read; tenants who rented the property write
POST <code>/reports</code> , GET <code>/reports/mine</code>	Auth (must have rented or bought the item)
GET <code>/reports</code> , GET <code>/reports/open-count</code> , GET <code>/reports/:id</code> , PATCH <code>/reports/:id/status transfer</code>	Admin (any)
GET POST <code>/threads</code> , POST <code>/threads/support</code> , GET <code>/threads/:id/summary messages</code> , POST <code>/threads/:id/messages</code> , PATCH <code>/threads/:id/read</code>	Auth (per-thread access rules)
GET <code>/threads/support-queue</code> , PATCH <code>/threads/:id/claim resolve transfer</code>	Admin (any, per-thread rules)
PATCH <code>/threads/:id/reassign</code>	Admin (Super Admin)
GET <code>/notifications</code> , GET <code>/notifications/unread-count</code> , PATCH <code>/notifications/:id/read</code> , PATCH <code>/notifications/read-all</code>	Auth
POST GET PATCH <code>/vendors/me</code>	Tenant or Vendor
GET <code>/marketplace/products</code> , GET <code>/marketplace/products/:id</code>	Public
GET <code>/marketplace/products/mine</code> , POST <code>/marketplace/products</code> , PATCH DELETE <code>/marketplace/products/:id</code>	Tenant or Vendor with an approved shop (owner)

POST /marketplace/orders , GET /marketplace/orders/mine vendor , PATCH /marketplace/orders/items/:itemId/status read ship , POST ../confirm-received , GET ../tracking	Auth (buyer or vendor, checked per item)
GET /paystack/banks , GET /paystack/resolve-account	Auth
POST /paystack/webhook	Paystack only (signature checked)
POST /uploads/sign	Auth
POST /admin/bootstrap	Public with x-admin-bootstrap-secret ; only while no admin exists
GET /admin/me , PATCH /admin/me/on-duty , GET /admin/stats activity-feed activity-log	Admin (any)
GET /admin/users , GET /admin/users/:id , PATCH /admin/users/:id , GET /admin/vendors , GET /admin/vendors/:id , GET /admin/properties , GET /admin/properties/:id , GET /admin/marketplace/products orders , GET /admin/marketplace/orders/:id , GET /admin/messages/attention-count	Admin (any)
PATCH /admin/users/:id/deactivate email password , DELETE /admin/users/:id , POST /admin/users/:id/send-password-reset , PATCH /admin/vendors/:id/approve reject suspend unsuspend , PATCH /admin/properties/:id/relist , DELETE /admin/properties/:id , DELETE /admin/marketplace/products/:id , GET /admin/admins/pending-count , POST /admin/admins/:id/reset-password/request confirm	Admin (Moderator)
GET /admin/admins , POST /admin/admins/request confirm , PATCH /admin/admins/:id/level two-factor , DELETE /admin/admins/:id , PATCH /admin/users/:id/disable-2fa , DELETE /admin/activity-log , GET /admin/chat-log	Admin (Super Admin)

14. Conventions for new code

- **Never trust the client.** Every permission is checked in a guard or service, even if the app already hides the button.
- **Validate bodies with DTOs** (`class-validator`). The global pipe rejects unknown fields.
- **Money goes through `PaymentsService` only;** notifications go through `NotificationsService` only.
- **Race-safe state changes:** use conditional `updateMany` ("only if still unclaimed") or transactions, not read-then-write.
- **Audit sensitive admin actions** with `ActivityLogService.log`.
- **Allow for deleted people:** optional relations can be null after an account is deleted.
- **Error messages are user-facing:** the app shows them as-is.
- **Comment the why:** the existing code explains reasons, not mechanics. Keep doing that.